Main Purpose

- Develop a **general framework for incorporating inter-patient physiological variability into continuous ICU monitoring models**, using ECG-based detection of hemodynamic instability as the testbed.
- The framework's core contribution is methodological: a principled way to build and honestly evaluate a *personalized* deviation-from-baseline signal, rather than a population-averaged one—applicable in principle to any continuously monitored ICU signal.

Research Problem

- ICU monitoring signals vary enormously across patients — the same physiological change can point in different directions for different individuals, so population-level thresholds and population-trained models systematically underperform or mislead.
- The standard fix — normalizing each patient's signal against their own personal baseline — is conceptually right but is usually implemented inconsistently across studies, with two recurring, undisclosed weaknesses:
  - the baseline is often built from the same data being evaluated (self-referential, inflates apparent performance), and
  - It's rarely checked against what a real-time bedside monitor could actually compute at the moment of scoring (deployability is assumed, not verified).
- No existing framework in this space explicitly separates and quantifies these two failure modes, meaning reported performance for "patient-calibrated" monitoring systems is often not directly interpretable as either a ceiling or a deployable estimate.

Background

- ICU hemodynamic monitoring currently relies on intermittent vital-sign sampling, introducing a 15–35 minute detection lag before instability is recognized.
- Continuous signals like ECG could close this gap, but only if patient-specific variability is handled correctly — otherwise, population-pooled statistical and ML approaches understate or overstate the true signal (as shown by the pseudoreplication/patient-level discrepancy in prior work).
- This makes the *evaluation methodology itself—not* just the choice of features or classifier—the actual bottleneck limiting real deployment of personalized ICU monitoring.

Research Solution: The Framework

- **A two-tier baseline-construction protocol** that any ICU monitoring model can adopt:
  - *Tier 1 (ceiling estimate)—jackknife-style* personalization: leave-one-observation-out baseline per patient, using all available data, answering "How much personalized signal exists at all?"

    - *Tier 2 (deployable estimate)* — calibration-window personalization: baseline built only from an initial observation period preceding everything evaluated, answering "how much of that signal survives the constraint of real-time computability."
- **A leakage-disciplined evaluation protocol** wrapped around both tiers: nested leave-one-subject-out validation, with feature selection performed strictly after normalization and strictly within each training fold—ensuring the reported gap between the two tiers reflects the baseline-construction choice alone, not incidental methodological leakage.
- **A model-agnostic validation layer**: benchmarking multiple classifier families (linear through nonlinear/ensemble) under both tiers to confirm findings reflect genuine physiological structure rather than one model's idiosyncrasy, plus calibration analysis (Brier score and reliability curves) to verify the resulting risk scores are usable, not just rank-ordering.
- **Applied here to ECG/hemodynamic instability in a 20-patient cardiogenic shock cohort** as a concrete demonstration, with the explicit claim that the two-tier protocol generalizes to other continuously monitored ICU signals facing the same personalization-vs-deployability tradeoff.